

Q2-B: In $G = \mathbb{Z}_{21}^* = \{1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20\}$ show that the set $H = \{1, 4, 16\}$ is a subgroup. Since G is abelian, H is automatically normal. Identify the four elements of G/H and construct their multiplication table. Is this group isomorphic to $\mathbb{Z}_4$?

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Setup

Let

$$G = \mathbb{Z}_{21}^* = \{1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20\}$$

be the multiplicative group of units modulo 21, with $|G| = \phi(21) = 12$, and let

$$H = \{1, 4, 16\}.$$

Part 1: H is a Subgroup of G

Proposition 1. $H = \{1, 4, 16\}$ is a subgroup of $G = \mathbb{Z}_{21}^*$.

Proof. We use the **Subgroup Test**: a non-empty subset H of a group G is a subgroup if and only if for all $a, b \in H$, the element $ab^{-1} \in H$.

It suffices to verify closure under multiplication and the existence of inverses within H .

Step 1: Find inverses of elements of H in G .

We compute inverses modulo 21:

$$1^{-1} \equiv 1 \pmod{21}, \quad 4^{-1} \equiv 16 \pmod{21} \text{ since } 4 \cdot 16 = 64 \equiv 1 \pmod{21},$$

$$16^{-1} \equiv 4 \pmod{21} \text{ since } 16 \cdot 4 = 64 \equiv 1 \pmod{21}.$$

All inverses lie in H . ✓

Step 2: Closure under multiplication.

We compute all products $a \cdot b \pmod{21}$ for $a, b \in H$:

$\cdot \pmod{21}$	1	4	16
1	1	4	16
4	4	16	$64 \equiv 1$
16	16	$64 \equiv 1$	$256 \equiv 4$

All products lie in H . ✓

Since H is non-empty, closed under multiplication, and closed under inverses, $H \leq G$. $\square$

Remark 1. Observe that $H = \{1, 4, 16\} = \langle 4 \rangle$, the cyclic subgroup generated by 4, since $4^1 = 4$, $4^2 = 16$, $4^3 = 64 \equiv 1 \pmod{21}$. Thus $|H| = 3$, which divides $|G| = 12$, consistent with Lagrange's Theorem.

Part 2: H is Normal in G

Proposition 2. $H \trianglelefteq G$.

Proof. Since $G = \mathbb{Z}_{21}^*$ is an **abelian group** (multiplication modulo 21 is commutative), every subgroup of G is normal. Indeed, for all $g \in G$ and $h \in H$:

$$ghg^{-1} = gg^{-1}h = h \in H.$$

Hence $gHg^{-1} = H$ for all $g \in G$, so $H \trianglelefteq G$. $\square$

Part 3: The Four Cosets of H in G

Since $|G| = 12$ and $|H| = 3$, the index $[G : H] = \frac{12}{3} = 4$, so G/H has exactly **four** cosets. We compute the left cosets $gH = \{g \cdot h \pmod{21} : h \in H\}$:

$$\begin{aligned} 1H: & 1 \cdot \{1, 4, 16\} = \{1, 4, 16\} \\ 2H: & 2 \cdot \{1, 4, 16\} = \{2, 8, 32 \bmod 21\} = \{2, 8, 11\} \\ 5H: & 5 \cdot \{1, 4, 16\} = \{5, 20, 80 \bmod 21\} = \{5, 20, 17\} \\ 10H: & 10 \cdot \{1, 4, 16\} = \{10, 40 \bmod 21, 160 \bmod 21\} = \{10, 19, 13\} \end{aligned}$$

Verification: The four cosets are pairwise disjoint and their union is:

$$\{1, 4, 16\} \cup \{2, 8, 11\} \cup \{5, 20, 17\} \cup \{10, 19, 13\} = G. \quad \checkmark$$

So the four elements of G/H are:

$$G/H = \{H, 2H, 5H, 10H\}.$$

Part 4: Multiplication Table of G/H

The group operation on cosets is defined by $(aH)(bH) = (ab)H$, computed modulo 21.

We compute each product of coset representatives and identify which coset the result falls in:

- $2 \cdot 5 = 10 \in 10H$, $2 \cdot 10 = 20 \in 5H$, $5 \cdot 10 = 50 \equiv 8 \pmod{21} \in 2H$
- $2 \cdot 2 = 4 \in H$, $5 \cdot 5 = 25 \equiv 4 \pmod{21} \in H$, $10 \cdot 10 = 100 \equiv 16 \pmod{21} \in H$

$\cdot$	H	$2H$	$5H$	$10H$
H	H	$2H$	$5H$	$10H$
$2H$	$2H$	H	$10H$	$5H$
$5H$	$5H$	$10H$	H	$2H$
$10H$	$10H$	$5H$	$2H$	H

The key observations from the table are:

- The identity element is H .
- $(2H)^2 = H$, $(5H)^2 = H$, $(10H)^2 = H$.
- Every non-identity element has order 2.

Part 5: Is $G/H \cong \mathbb{Z}_4$?

Theorem 1. $G/H \not\cong \mathbb{Z}_4$. Instead, $G/H \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ (the Klein four-group V_4).

Proof. Recall that $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ under addition modulo 4. It is a **cyclic** group of order 4, containing an element of order 4.

From the multiplication table of G/H , the orders of elements are:

$$\text{ord}(H) = 1, \quad \text{ord}(2H) = 2, \quad \text{ord}(5H) = 2, \quad \text{ord}(10H) = 2.$$

Since **every non-identity element of G/H has order 2**, the group G/H contains no element of order 4. But $\mathbb{Z}_4$ has an element of order 4 (namely, the generator $1 \in \mathbb{Z}_4$).

Since $\mathbb{Z}_4$ and G/H have different element orders, they cannot be isomorphic:

$$G/H \not\cong \mathbb{Z}_4.$$

Moreover, G/H is an abelian group of order 4 in which every non-identity element has order 2. Up to isomorphism, the only such group is the **Klein four-group**:

$$V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0, 0), (1, 0), (0, 1), (1, 1)\}.$$

The isomorphism $\varphi : G/H \rightarrow \mathbb{Z}_2 \times \mathbb{Z}_2$ is given explicitly by:

$$\varphi(H) = (0, 0), \quad \varphi(2H) = (1, 0), \quad \varphi(5H) = (0, 1), \quad \varphi(10H) = (1, 1).$$

One can verify that φ preserves the group operation. For example:

$$\varphi(2H \cdot 5H) = \varphi(10H) = (1, 1) = (1, 0) + (0, 1) = \varphi(2H) + \varphi(5H). \quad \checkmark$$

Hence,

$$\boxed{G/H \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \quad G/H \not\cong \mathbb{Z}_4.}$$

□